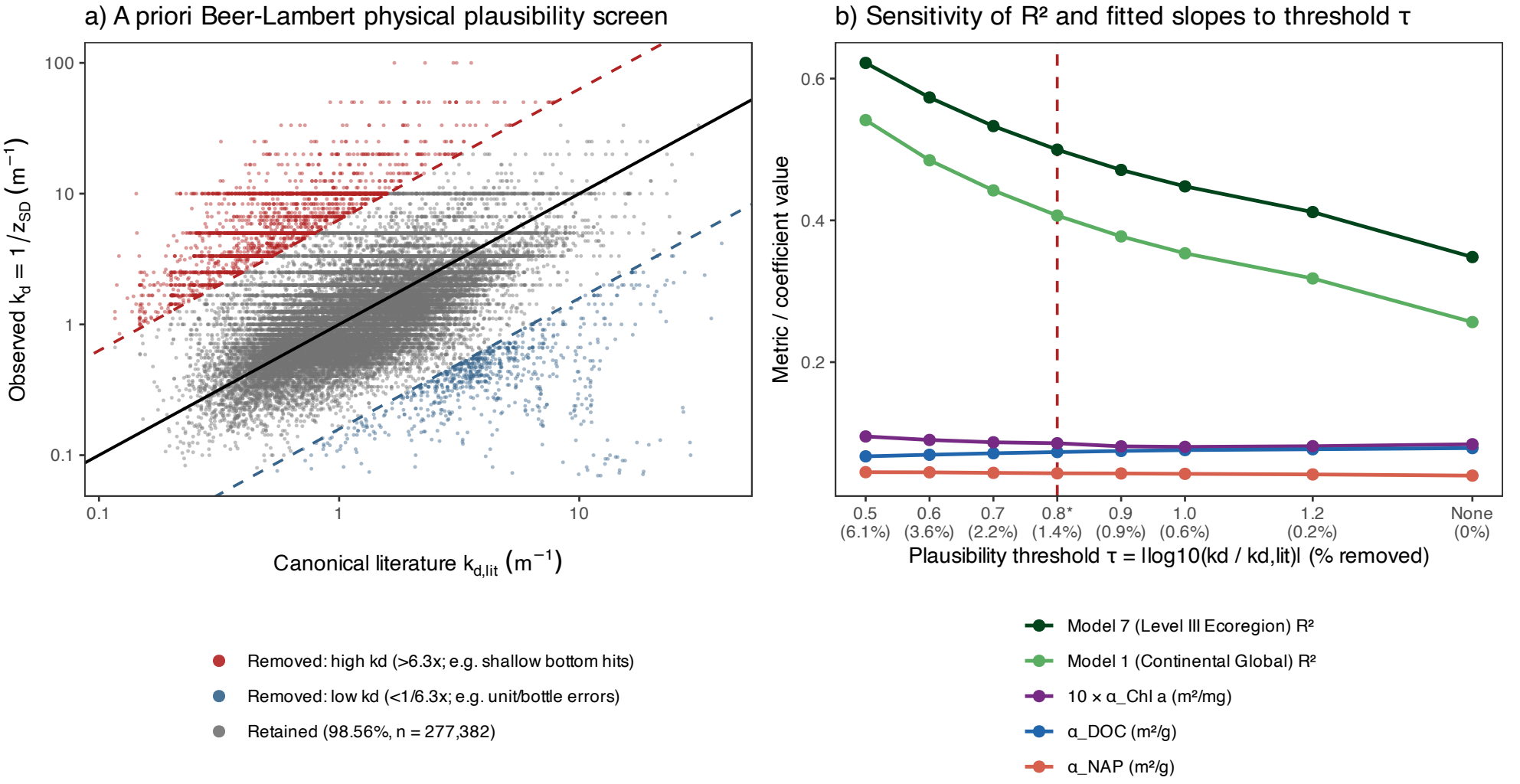

a) A priori Beer-Lambert physical plausibility screen
Observed $k_d = 1/z_{SD}$ ($m^{-1}$)
100
10
1
0.1
0.1
1
10
Canonical literature $k_{d,lit}$ ($m^{-1}$)
Removed: high kd (>6.3x; e.g. shallow bottom hits)
Removed: low kd (<1/6.3x; e.g. unit/bottle errors)
Retained (98.56%, n = 277,382)
b) Sensitivity of R² and fitted slopes to threshold τ
Metric / coefficient value
0.6
0.4
0.2
0.5 (6.1%)
0.6 (3.6%)
0.7 (2.2%)
0.8* (1.4%)
0.9 (0.9%)
1.0 (0.6%)
1.2 (0.2%)
None (0%)
Plausibility threshold τ = |log10(kd / kd,lit)| (% removed)
Model 7 (Level III Ecoregion) R²
Model 1 (Continental Global) R²
10 × α_Chl a (m²/mg)
α_DOC (m²/g)
α_NAP (m²/g)